



Student Name: _____

Teacher: _____

Sydney Technical High School

2025

PRELIMINARY
YEARLY
EXAMINATION

Yr 11 Mathematics Extension 1

General Instructions

- Working Time – 90 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section 2, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless work or illegible writing
- **Begin each question on a new page**

Total marks:
65

Section 1 — 5 marks

- Attempt Questions 1 - 5
- Allow about 10 minutes for this part

Section 2 — 60 marks

- Attempt Questions 6 -11
- Allow about 80 minutes for this section.

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SECTION 1

- 5 marks
- Attempt Questions 1-5.
- Allow about 10 mins for this section.
- Use the multiple-choice answer sheet for Questions 1-5.

Marks

Question 1

Which is equivalent to $\ln e^{x^2}$?

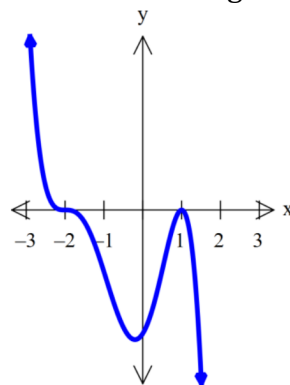
1

- A. e^{2x}
- B. $2e^x$
- C. $2x$
- D. x^2

Question 2

Which equation best represents the following?

1



- A. $y = (x - 1)^2(x + 2)^3$
- B. $y = -(x - 1)^2(x + 2)^3$
- C. $y = (x + 1)^2(x - 2)^3$
- D. $y = -(x + 1)^2(x - 2)^3$

Question 3

Which of the following is a factor of $6x^3 - 5x^2 - 34x - 15$?

1

- A. $x + 1$
- B. $x - 3$
- C. $x - 2$
- D. $x + 5$

Question 4

A particle is moving along the x - axis.

1

The displacement (x metres) of a particle from the origin at a time (t seconds) is given by the equation:

$$x = 6t^2 - \frac{12}{t}$$

What is the acceleration of the particle after 2 seconds?

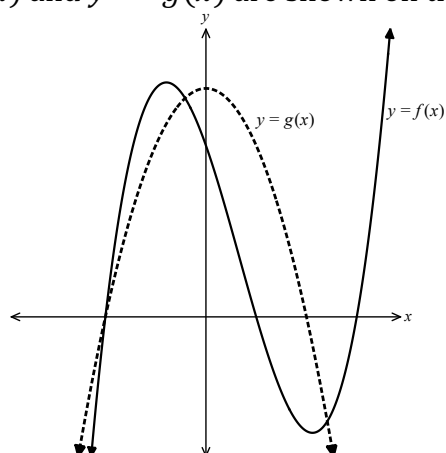
- A. 9 ms^{-2}
- B. 18 ms^{-2}
- C. 21 ms^{-2}
- D. 24 ms^{-2}

Multiple Choice Questions continue on the next page.

Question 5

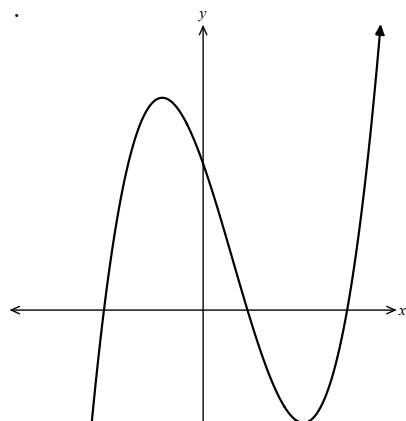
1

The graphs of $y = f(x)$ and $y = g(x)$ are shown on the set of axes below.

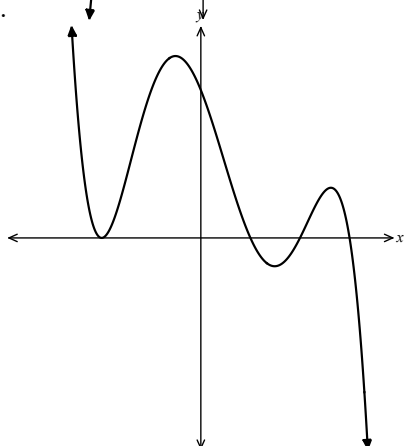


Which graph below could represent $y = f(x) \times g(x)$?
(Scales differ on the y -axes)

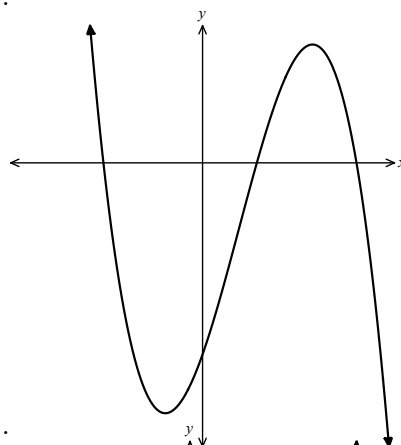
A. .



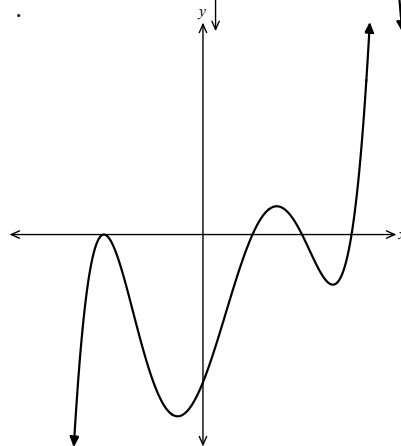
B. .



C. .



D. .



SECTION 2

- 60 marks
- Attempt Questions 6-11.
- Allow about 80 mins for this section.
- Start your answer for each question on a new page in the answer booklet.
- In Questions 6 – 11, your responses should include all relevant mathematical reasoning and/or calculations.

Question 6 (10 marks) **Start a NEW page**

Marks

- a) (i) Show that $(x + 3)$ is a factor of $x^3 - 3x^2 - 10x + 24$ 1
- (ii) Hence, fully factorise $x^3 - 3x^2 - 10x + 24$ 2
- (iii) Sketch the polynomial $P(x) = x^3 - 3x^2 - 10x + 24$ 2
- b) Professor Snape has a colony of bacteria. Initially, there are 1000 bacteria.
The number of bacteria, $N(t)$, after t minutes, is given by:
- $$N(t) = 1000e^{kt}$$
- (i) After 20 minutes there are 2000 bacteria. Show that $k = \frac{1}{20} \ln 2$ 1
- (ii) How many bacteria are there after 120 minutes? 1
- (iii) What is the rate of change of the number of bacteria per minute when $t = 120$? 1
- (iv) How long does it take for the number of bacteria to increase from 1000 to 100 000? 2

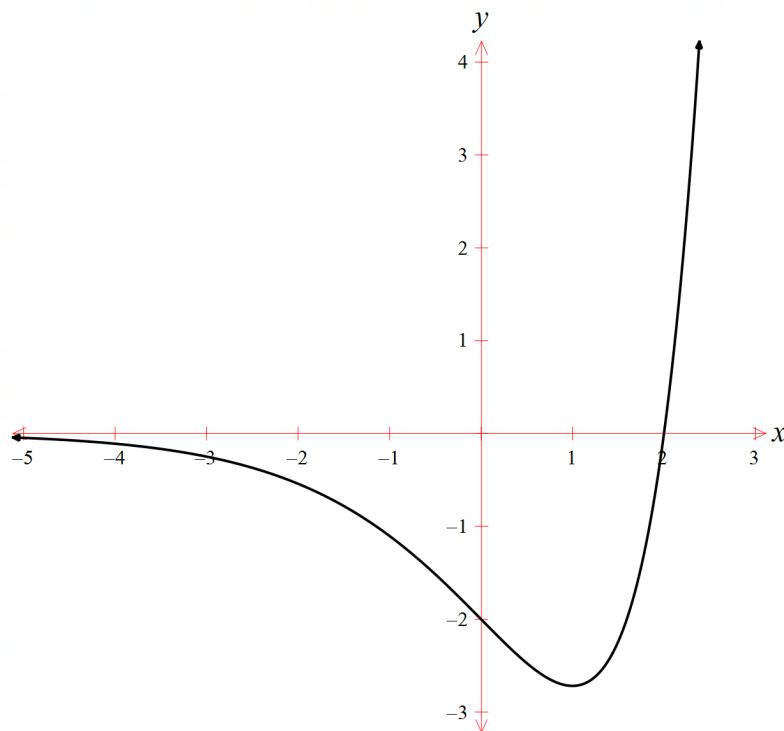
Question 7 (10 marks) **Start a NEW page**

Marks

- a) A rectangle of length l cm has a fixed area of 48 cm^2 . If the length of the rectangle is increasing at a rate of 5 cm/sec , find,
- (i) the rate at which the breadth of the rectangle is decreasing when the length of the rectangle is 8 cm 2
- (ii) the rate of change of the perimeter, when the length is 8 cm . 2
- b) An odd polynomial of degree 5 has zeros of -1 and 3 .
- Find the expanded equation of the polynomial if it also passes through the point $(2, 60)$ 3
- c) Solve the for x , $(\tan x + 1)(\sin 2x - 1) = 0$ for $0 \leq x \leq 2\pi$ 3

For all graphing questions, please use the grids provided in the answer booklet.

- a) The diagram below shows the graph of the function $y = f(x)$



Sketch showing all important features:

(i) $y = |f(x)|$ 2

(ii) $y = \sqrt{f(x)}$ 2

- b) Find the cartesian equation of the curve whose parametric equations are:

$$x = 1 + 2^{-t} \quad \text{and} \quad y = 2^t \quad \text{3}$$

- c) Find the equation of the normal to the curve with parametric equation:

$$x = 1 + t^2 \quad \text{and} \quad y = \frac{1-t}{1+t} \quad \text{at the point where } t = 1. \quad \text{3}$$

Question 9 (10 marks) **Start a NEW page**

Marks

- a) A polynomial, $P(x)$, has a remainder of $(3x + 5)$ when it is divided by $2x^2 - 3x - 2$.

2

What is the remainder when it is divided by $(x - 2)$?

- b) If α, β, γ are roots of the cubic equation $2x^3 - 4x^2 - 6x + 8 = 0$.

Find the value of:

(i) $\alpha + \beta + \gamma$

1

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$

1

(iii) $\alpha^2 + \beta^2 + \gamma^2$

2

(iv) $\alpha^3 + \beta^3 + \gamma^3$

2

c) solve $\frac{x^2 - 4}{x} > 0$

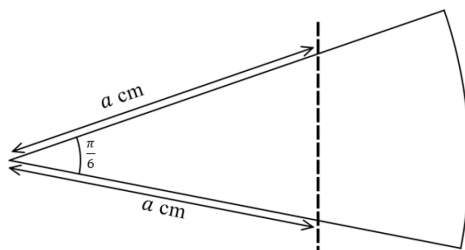
2

Question 10 (10 marks) **Start a NEW page**

Marks

- a) Kevin buys a slice of cake in the shape of a sector.
The radius is r cm and the sector angle is $\frac{\pi}{6}$ radians.
He wants to give half of the slice to his friend Michael. He makes a cut
across the sector as shown, a cm away from the centre.

2



Show that when they each have half the slice, $a = r \sqrt{\frac{\pi}{6}}$

- b) A particle is moving along the x -axis. Its velocity v at time t is given by

2

$$v = \sqrt{20t - 2t^2} \text{ metres per second}$$

Find the acceleration of the particle when $t = 4$.

Express your answer as an exact value in its simplest form.

- c) The equation $4x^3 - 27x + k = 0$ has a double root.

2

Find the possible value(s) of k .

- d) Consider the cubic curve $y = x^3 - 2x$

- (i) Write down the equation of the tangent to the curve at the
point where $x = a$, in terms of a .

2

- (ii) Find, in terms of a , the x co-ordinate of the point where this
tangent meets the curve again.

2

Question 11 (10 marks) **Start a NEW page**

Marks

- a) A frittata, when it is removed from the oven, has an initial temperature of 180°C .

It is placed in a room where its surrounding temperature is 26°C .

It cools so that its temperature $T(\text{in } ^{\circ}\text{C})$ satisfies the equation

$$\frac{dT}{dt} = -k(T - 26), \quad k > 0$$

where t is the time in minutes since the frittata was removed from the oven.

- | | | |
|-------|--|---|
| (i) | Show that $T = 26 + Ae^{-kt}$ satisfies this equation. | 1 |
| (ii) | Fifteen minutes after being removed from the oven, the temperature of the frittata is 150°C . What is its temperature after a further fifteen minutes? | 2 |
| | | |
| b) | A function is defined by $f(x) = 4 - \left(1 - \frac{x}{3}\right)^2$ over the domain $(-\infty, 3]$ | |
| (i) | Sketch the graph of $y = f(x)$ showing the x - and y -intercepts. | 2 |
| (ii) | Find the equation of the inverse function $f^{-1}(x)$ and state its domain. | 3 |
| (iii) | Find the point of intersection of $f(x)$ and $f^{-1}(x)$ leaving your answer in surd form. | 2 |

END OF EXAMINATION

MULTIPLE CHOICE ANSWER SHEET

Completely fill the response oval representing the most correct answer.

Do not remove this sheet.

1. A ☐ B ☐ C ☐ D ☒
2. A ☐ B ☒ C ☐ D ☐
3. A ☐ B ☒ C ☐ D ☐
4. A ☒ B ☐ C ☐ D ☐
5. A ☐ B ☒ C ☐ D ☐

End of Section 1
Please turn to the next page for Section 2

Yr 11 Ex1 yearly 2025 solutions

Question 6

a) i) $P(x) = x^3 - 3x^2 - 10x + 24$

$$\begin{aligned} P(-3) &= (-3)^3 - 3(-3)^2 - 10(-3) + 24 \\ &= -27 - 27 + 30 + 24 \\ &= 0 \end{aligned}$$

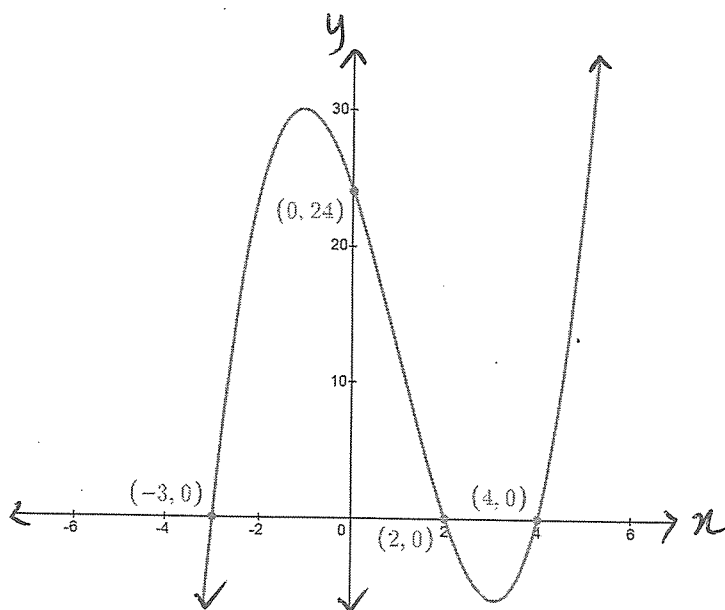
$\therefore (x+3)$ is a factor. (1)

ii)

$$\begin{array}{r} x^2 - 6x + 8 \\ x+3 \overline{) x^3 - 3x^2 - 10x + 24} \\ \underline{x^3 + 3x^2} \\ -6x^2 - 10x \\ \underline{-6x^2 - 18x} \\ 8x + 24 \\ \underline{8x + 24} \\ 0 \end{array}$$

$$\begin{aligned} \therefore P(x) &= (x+3)(x^2 - 6x + 8) \\ &= (x+3)(x-4)(x-2) \end{aligned} \quad (2)$$

iii)



(2)

Q6 b) i) $N(t) = 1000e^{kt}$

Given $N(t) = 2000$ when $t = 20$

$$2000 = 1000e^{20k}$$

$$e^{20k} = 2$$

$$\ln(e^{20k}) = \ln 2$$

$$20k = \ln 2$$

$$k = \frac{1}{20} \ln 2$$

(1)

ii) when $t = 120$

$$N(t) = 1000e^{120 \times \frac{\ln 2}{20}}$$

$$= 64000 \text{ bacteria}$$

(1)

OR

$$N(t) = 1000e^{120(0.0347)}$$

$$= 64328 \text{ bacteria}$$

iii) $N' = kN$

$$= \frac{\ln 2}{20} \times 64000$$

$$= 2218 \text{ bacteria/min}$$

$$N' = kN$$

$$= 0.0347 \times 64328$$

$$= 2232 \text{ bacteria/min}$$

(1)

Q6 iv) $N(t) = 1000 e^{\frac{\ln 2}{20} t}$

given $N(t) = 100000$

$$100000 = 1000 e^{\frac{\ln 2}{20} t}$$

$$e^{\frac{\ln 2}{20} t} = 100$$

$$\ln(e^{\frac{\ln 2}{20} t}) = \ln 100$$

$$\frac{\ln 2}{20} t = \ln 100$$

$$t = \ln 100 \div \frac{\ln 2}{20}$$

$$\doteq 132.877, \dots$$

$$\doteq 133 \text{ minutes}$$

(2)

Question 7

a) i) Find $\frac{db}{dt}$ given: $\frac{dl}{dt} = 5$

and $A = bl = 48$

• $bl = 48$

$$b = \frac{48}{l}$$

$$= 48l^{-1}$$

$$\frac{db}{dl} = -48l^{-2}$$

$$= -\frac{48}{l^2}$$

$$\therefore \frac{db}{dt} = \frac{db}{dl} \times \frac{dl}{dt}$$

$$\Rightarrow \text{length} = 8 \text{ cm}$$

$$\therefore \frac{db}{dt} = \frac{-48}{8^2} \times 5$$

$$= -3.75 \text{ cm/s} \quad \textcircled{2}$$

ii) when $l = 8$ $b = \frac{48}{8} = 6$

Also $P = 2(l+b)$

$$\therefore \frac{dP}{dt} = 2 \left[\frac{dl}{dt} + \frac{db}{dt} \right]$$

$$= 2 [5 - 3.75]$$

$$= 2.5 \text{ cm/s}$$

$$\textcircled{2}$$

Q7b) Note: an odd polynomial means its function satisfies:

$$f(-x) = -f(x)$$

so the graph is symmetric about the origin.

Also degree of 5 means leading term is ax^5 .

• zeros at $x = -1$ and $x = 3$

$\therefore$ factors of $(x+1)$ and $(x-3)$

Also $x = 1$ and $x = -3$ are also factors with the same multiplicity.

and $x = 0$ as we need degree 5.

$$\therefore f(x) = a(x+1)(x-1)(x+3)(x-3)x$$

$$= ax(x^2-1)(x^2-9)$$

$$= ax(x^4 - 9x^2 - x^2 + 9)$$

$$= ax(x^4 - 10x^2 + 9)$$

$$= a(x^5 - 10x^3 + 9x)$$

Since passes thru: $(2, 60)$

$$\begin{aligned} f(2) &= a(2^5 - 10(2)^3 + 9 \times 2) \\ &= a(-30) \end{aligned}$$

Q7b) Cont.

$$-30a = 60$$

$$a = \frac{60}{-30}$$

$$\therefore a = -2$$

$$\therefore f(x) = -2(x^5 - 10x^3 + 9x)$$

$$= -2x^5 + 20x^3 - 18x \quad (3)$$

c) $(\tan x + 1)(\sin 2x - 1) = 0$

• When $\tan x + 1 = 0$

$$\tan x = -1$$

$$\therefore x = -\frac{\pi}{4}$$

$\therefore$ 2nd + 4th quadrant

$$\therefore x = \pi - \frac{\pi}{4}$$

$$= \frac{3\pi}{4}$$

OR

$$x = 2\pi - \frac{\pi}{4}$$

$$= \frac{7\pi}{4}$$

• when $\sin 2x - 1 = 0$

$$\sin 2x = 1$$

$$\therefore 2x = \frac{\pi}{2}$$

$$\therefore x = \frac{\pi}{4}$$

$$2x = \frac{\pi}{2} + 2\pi$$

$$2x = \frac{5\pi}{2}$$

$$\therefore x = \frac{5\pi}{4}$$

[note: can not be $\frac{9\pi}{4}$ as

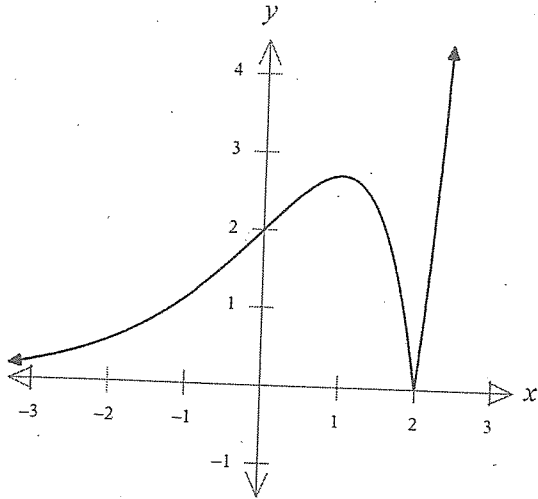
outside domain

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

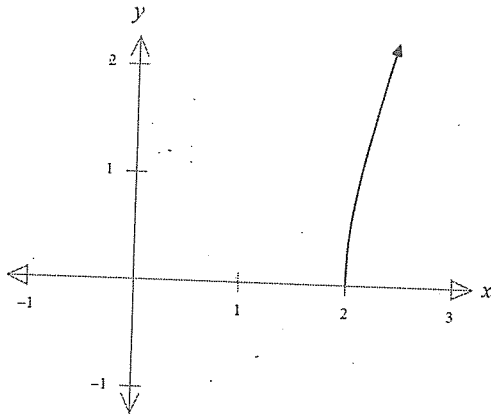
Question 8

a)

(i) $y = |f(x)|$



(ii) $y = \sqrt{f(x)}$



Q8 Cont.

b) $x = 1 + 2^{-t}$

$$x - 1 = 2^{-t}$$

$$(x-1)^{-1} = (2^{-t})^{-1}$$

$$\frac{1}{x-1} = 2^t$$

given $y = 2^t$

$$\therefore y = \frac{1}{x-1}$$

c) $x = 1 + t^2$

$$\frac{dx}{dt} = 2t$$

$$y = \frac{1-t}{1+t}$$

$$u = 1-t$$

$$u' = -1$$

$$v = 1+t$$

$$v' = 1$$

$$\frac{dy}{dt} = \frac{vu' - uv'}{v^2}$$

$$= \frac{-1(1+t) - 1(1-t)}{(1+t)^2}$$

$$= \frac{-1-t-1+t}{(1+t)^2}$$

$$= \frac{-2}{(1+t)^2}$$

• when $t = 1$

$$x = 1 + (1)^2$$

$$= 2$$

$$y = \frac{1-1}{1+1} = \frac{0}{2} = 0$$

$$\therefore \text{pt} \Rightarrow (2, 0)$$

$$\therefore \text{gradient} \Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{-2}{(1+t)^2} \times \frac{1}{2t}$$

Q8 c) Cont.

$$\text{at } t=1 \quad \frac{dy}{dx} = \frac{-2}{(1+1)^2} \times \frac{1}{2 \times 1}$$

$$= \frac{-2}{2^2} \times \frac{1}{2}$$

$$= \frac{-2}{4} \times \frac{1}{2}$$

$$= \frac{-2}{8}$$

$$= \frac{-1}{4}$$

$\therefore$ gradient of normal = 4

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 4(x - 2)$$

$$y = 4x - 8$$

Question 9

$$\begin{aligned} \text{a)} \quad P(x) &= (2x^2 - 3x - 2)Q(x) + (3x + 5) \\ &= (2x + 1)(x - 2)Q(x) + (3x + 5) \end{aligned}$$

$$\begin{aligned} P(2) &= (2 \times 2 + 1)(2 - 2)Q(x) + (3 \times 2 + 5) \\ &= 5 \times 0 \times Q(x) + 11 \\ &= 0 + 11 \\ &= 11 \end{aligned}$$

$\therefore$ the remainder is 11. 2

$$\text{b)} \quad 2x^3 - 4x^2 - 6x + 8 = 0$$

$$\begin{aligned} \text{i)} \quad x + B + y &= \frac{-b}{a} \\ &= \frac{-(-4)}{2} \\ &= 2 \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad xB + x y + B y &= \frac{c}{a} = \frac{-6}{2} \\ &= -3 \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{iii)} \quad x^2 + B^2 + y^2 &= (x + B + y)^2 - 2(xB + x y + B y) \\ &= (2)^2 - 2(-3) \\ &= 4 + 6 \\ &= 10 \quad \text{②} \end{aligned}$$

Q9 Cont.

iv) $\alpha^3 + \beta^3 + \gamma^3$

→ as α, β, γ are roots, they satisfy the polynomial.

$$\therefore 2\alpha^3 = 4\alpha^2 + 6\alpha - 8$$

$$2\beta^3 = 4\beta^2 + 6\beta - 8$$

$$2\gamma^3 = 4\gamma^2 + 6\gamma - 8$$

$$\text{thus } 2\alpha^3 + 2\beta^3 + 2\gamma^3 = 4\alpha^2 + 4\beta^2 + 4\gamma^2 + 6\alpha + 6\beta + 6\gamma - 24$$

$$2(\alpha^3 + \beta^3 + \gamma^3) = 4(\alpha^2 + \beta^2 + \gamma^2) + 6(\alpha + \beta + \gamma) - 24$$

$$= 4 \times 10 + 6 \times 2 - 24$$

$$= 40 + 12 - 24$$

$$2(\alpha^3 + \beta^3 + \gamma^3) = 28$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = \frac{28}{2}$$

$$= 14$$

(2)

c) $\frac{x^2 - 4}{x} > 0$

$$\frac{(x+2)(x-2)}{x} > 0$$

Cont.

Q9 Cont.

$$c) \frac{x^2 - 4}{x} > 0$$

$$\frac{(x+2)(x-2)}{x} > 0$$

∴ critical points when $x = -2$, $x = 0$, $x = 2$

test

$$\bullet \quad x < -2 \quad \text{test } x = -3 \quad \begin{array}{c} \text{Sign} \\ \frac{(-)(-)}{-} = - < 0 \end{array}$$

$$\bullet \quad -2 < x < 0 \quad \text{test } x = -1 \quad \frac{(+)(-)}{-} = + > 0$$

$$\bullet \quad 0 < x < 2 \quad \text{test } x = 1 \quad \frac{(+)(-)}{+} = - < 0$$

$$\bullet \quad x > 2 \quad \text{test } x = 3 \quad \frac{(+)(+)}{+} = + > 0$$

$$\bullet \quad -2 < x < 0 \quad \text{OR} \quad x > 2$$

OR

$$(-2, 0) \cup (2, \infty)$$

Question 10

a) Area of triangle

$$A = \frac{1}{2}ab\sin C$$

$$= \frac{1}{2} \times a^2 \times \sin \frac{\pi}{6}$$

$$= \frac{1}{2} \times a^2 \times \frac{1}{2}$$

$$= \frac{a^2}{4}$$

Area of Sector

$$A = \frac{1}{2}r^2\theta$$

$$= \frac{1}{2}r^2 \times \frac{\pi}{6}$$

$$= \frac{\pi r^2}{12}$$

• When is area of triangle equal to $\frac{1}{2}$ area of the sector

$$\frac{a^2}{4} = \frac{1}{2} \times \frac{\pi r^2}{12}$$

$$\frac{a^2}{4} = \frac{\pi r^2}{24}$$

$$a^2 = \frac{4\pi r^2}{24}$$

$$a^2 = \frac{\pi r^2}{6}$$

Since a is length
thus $a > 0$

$$\therefore a = +\sqrt{\frac{\pi r^2}{6}}$$

$$\therefore a = r\sqrt{\frac{\pi}{6}}$$

$$a = \pm r\sqrt{\frac{\pi}{6}}$$

Question 10 Cont.

$$b) \quad v = \sqrt{20t - 2t^2} = (20t - 2t^2)^{\frac{1}{2}}$$

$$a = \frac{dv}{dt}$$

$$= \frac{1}{2} \times (20t - 2t^2)^{-\frac{1}{2}} \times (20 - 4t)$$

$$= \frac{10 - 2t}{\sqrt{20t - 2t^2}}$$

When $t = 4$

$$a = \frac{10 - 2 \times 4}{\sqrt{20 \times 4 - 2(4)^2}}$$

$$= \frac{10 - 8}{\sqrt{80 - 32}}$$

$$= \frac{2}{\sqrt{48}}$$

$$= \frac{2}{4\sqrt{3}}$$

$$= \frac{1}{2\sqrt{3}}$$

$$= \frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{6}$$

Q10 cont.

c) $4x^3 - 27x + k = 0$

$$\frac{d}{dx} (4x^3 - 27x + k) = 12x^2 - 27$$

$$= 3(4x^2 - 9)$$

$$= 3(2x-3)(2x+3)$$

$$\therefore x = \frac{3}{2} \text{ or } x = -\frac{3}{2}$$

• For $x = \frac{3}{2}$

$$4\left(\frac{3}{2}\right)^2 - 27\left(\frac{3}{2}\right) + k = 0$$

$$\therefore k = 27$$

• For $x = -\frac{3}{2}$

$$4\left(-\frac{3}{2}\right)^2 - 27\left(-\frac{3}{2}\right) + k = 0$$

$$\therefore k = -27$$

Q10 Cont.

d) $y = x^3 - 2x$ when $x = a$

i) when $x = a$
 $y = a^3 - 2a$

$$M_{\text{tangent}} = y' = 3x^2 - 2$$

at $x = a$ $m = 3a^2 - 2$

∴ eqn of tangent @ $x = a$

pt $(a, a^3 - 2a)$ $m = 3a^2 - 2$

$$y - y_1 = m(x - x_1)$$

$$y - (a^3 - 2a) = (3a^2 - 2)(x - a)$$

$$y - a^3 + 2a = x(3a^2 - 2) - a(3a^2 - 2)$$

$$y = x(3a^2 - 2) - 3a^3 + 2a + a^3 - 2a$$

$$\boxed{y = x(3a^2 - 2) - 2a^3}$$

ii) Cont.

Q10 Cont.

ii) Curve: $y = x^3 - 2x$ — (1)
tangent: $y = x(3a^2 - 2) - 2a^3$

• equating (1) = (2)

$$x^3 - 2x = x(3a^2 - 2) - 2a^3$$

$$x^3 - 2x - x(3a^2 - 2) + 2a^3 = 0$$

Simplifying:

$$x^3 - 2x - 3a^2x + 2x + 2a^3 = 0$$

$$\therefore x^3 - 3a^2x + 2a^3 = 0$$

as $x = a$ is a root $\Rightarrow (x - a)$ is a factor of the cubic

$\therefore x^3 - 3a^2x + 2a^3$ can be divided by $(x - a)$

Using the factor theorem:

$$x^3 - 3a^2x + 2a^3 = (x - a)(Ax^2 + Bx + C)$$

expanding:

$$(x - a)(Ax^2 + Bx + C) = Ax^3 + Bx^2 + Cx - Aax^2 - aBx - aC$$

$$= Ax^3 + (B - Aa)x^2 + (C - aB)x - aC$$

$$\therefore \bullet x^3 = Ax^3$$

$$A = 1$$

$$\bullet C - aB = -3a^2$$

$$C - a \times a = -3a^2$$

$$C - a^2 = -3a^2$$

$$C = -2a^2$$

$$\bullet B - Aa = 0$$

$$B - a = 0$$

$$B = a$$

$$\bullet -aC = 2a^3$$

$$-a \times -2a^2 = 2a^3 \checkmark$$

Cont.

pg 17

(Q10 cont. d) ii)

$$\therefore x^3 - 3a^2x + 2a^3 = (x-a)(Ax^2+Bx+C)$$
$$= (x-a)(x^2+ax-2a^2)$$

Solving $(x^2+ax-2a^2)$ using the quadratic formula

$$x = \frac{-a \pm \sqrt{a^2 - 4(1)(-2a^2)}}{2}$$
$$= \frac{-a \pm \sqrt{a^2 + 8a^2}}{2}$$
$$= \frac{-a \pm \sqrt{9a^2}}{2}$$
$$= \frac{-a \pm 3a}{2}$$

$$\therefore x = \frac{-a+3a}{2} \quad \text{OR} \quad x = \frac{-a-3a}{2}$$

$$= \frac{2a}{2}$$

$$\boxed{x = a}$$

$$x = \frac{-4a}{2}$$

$$\boxed{x = -2a}$$

$\therefore$ the tangent meets the curve again at:

$$x = -2a$$

Question 11

a) i) $T = 26 + Ae^{-kt} \Rightarrow Ae^{-kt} = T - 26$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$\therefore \frac{dT}{dt} = -k(T - 26)$$

ii) $T = 26 + Ae^{-kt}$

• When $t = 0$, $T = 180$

$$180 = 26 + Ae^0$$

$$180 = 26 + A$$

$$\therefore A = 180 - 26$$

$$A = 154$$

• When $t = 15$ $T = 150$

$$150 = 26 + 154e^{-15k}$$

$$124 = 154e^{-15k}$$

$$e^{-15k} = \frac{124}{154}$$

$$-15k = \ln\left(\frac{124}{154}\right)$$

$$k = \ln\left(\frac{124}{154}\right) \div -15$$

$$\therefore k = 0.01444473578, \dots$$

• When $t = 30$

$$T = 26 + 154e^{-0.01444473578 \times 30}$$

$$T = 125.84415584, \dots$$

$$T = 126^\circ \text{ (nearest degree)}$$

Q11 cont.

b) i) • y-intercept (let $x=0$)

$$\begin{aligned}f(0) &= 4 - [1 - 0]^2 \\&= 4 - 1 \\&= 3\end{aligned}$$

∴ y-intercept at $(0, 3)$

• x-intercepts Solve $f(x) = 0$

$$4 - \left(1 - \frac{x}{3}\right)^2 = 0$$

$$\left(1 - \frac{x}{3}\right)^2 = 4$$

$$1 - \frac{x}{3} = \pm 2$$

$$1 - \frac{x}{3} = 2 \quad \text{OR} \quad 1 - \frac{x}{3} = -2$$

$$-\frac{x}{3} = 1$$

$$-x = 3$$

$$x = -3$$

$$-\frac{x}{3} = -3$$

$$-x = -9$$

$$x = 9$$

outside domain
 $(-\infty, 3]$

∴ only valid x-intercept is

$$(-3, 0)$$

- endpoint at $x=3$

$$f(3) = 4 - \left(1 - \frac{3}{3}\right)^2$$

$$= 4 - 0$$

$$= 4$$

∴ endpoint at : $(3, 4)$ (note also max/vertex)

- In summary :

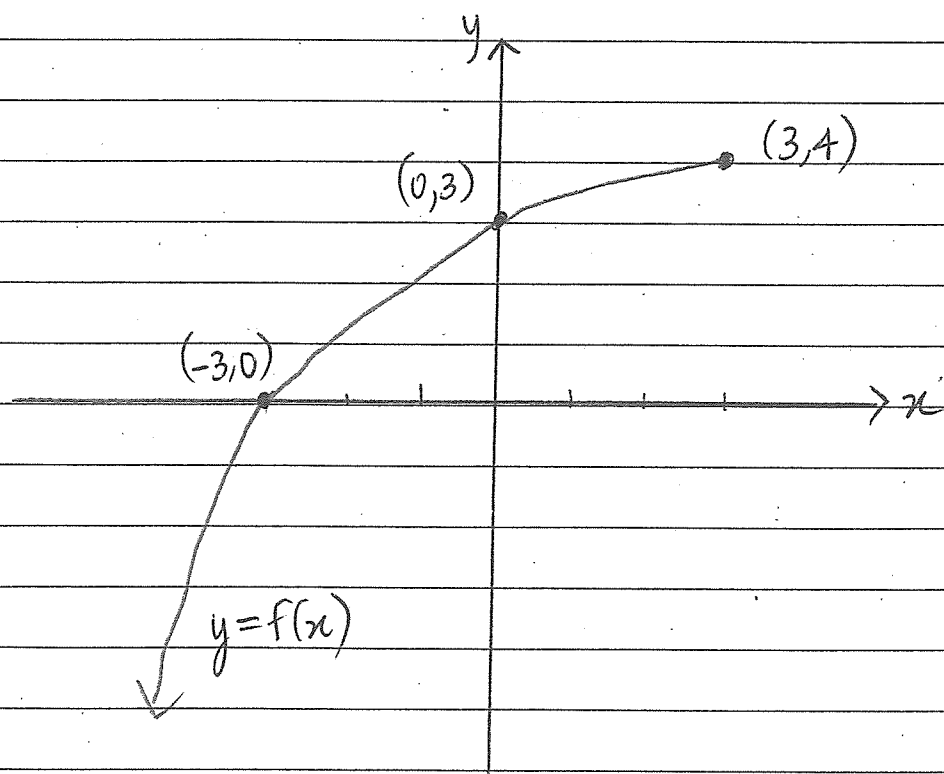
→ Parabola → concave down

→ endpt/vertex $(3, 4)$

→ x-int $(-3, 0)$

→ y-int $(0, 3)$

→ domain $(-\infty, 3]$ closed circle @ $x=3$



Q11 b) ii) $\rightarrow$ swap x & y

$$x = 4 - \left(1 - \frac{y}{3}\right)^2$$

$$\left(1 - \frac{y}{3}\right)^2 = 4 - x$$

$$1 - \frac{y}{3} = \pm \sqrt{4 - x}$$

$$-\frac{y}{3} = -1 \pm \sqrt{4 - x}$$

$$y = -3(-1 \pm \sqrt{4 - x})$$

$$y = 3(1 \mp \sqrt{4 - x})$$

• Range of $f(x)$ is $(-\infty, 4]$ so the domain of $f^{-1}(x)$ is $(-\infty, 4]$.

$$\therefore f^{-1}(x) = 3 - 3\sqrt{4 - x} \quad \text{for } x \text{ in the domain } (-\infty, 4]$$

Q11 b) iii).

$$f(x) = 4 - \left(1 - \frac{x}{3}\right)^2$$

pt of intersection occurs when:

$$f(x) = f^{-1}(x) \Rightarrow x = f(x)$$

$$\therefore x = 4 - \left(1 - \frac{x}{3}\right)^2$$

$$x = 4 - \left(1 - \frac{2x}{3} + \frac{x^2}{9}\right)$$

$$x = 3 + \frac{2x}{3} - \frac{x^2}{9}$$

$$9x = 27 + 6x - x^2$$

$$x^2 + 3x - 27 = 0$$

Solving using the quadratic formula:

$$x = \frac{-3 \pm \sqrt{3^2 - 4(1)(-27)}}{2}$$

$$= \frac{-3 \pm \sqrt{9 + 108}}{2}$$

$$= \frac{-3 \pm \sqrt{117}}{2}$$

$$\therefore x = \frac{-3 + \sqrt{117}}{2}$$

$$\text{OR } x = \frac{-3 - \sqrt{117}}{2}$$

$$x \doteq 3.9$$

not a possible
solution (outside
domain)

$$x \doteq -6.9$$

valid soln. ✓

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Question 6**a)**

- i. Mostly done well. Some students wasted time and used long division. Just use the factor theorem!
- ii. Mostly done well. Some simple mistakes with long division and inspection.
- iii. Mostly done well. Don't forget the y -intercept when sketching. No need to mark the coordinates of turning points unless you are asked to do so.

b)

- i. Mostly done well. Some careless mistakes.
- ii. Mostly done well. Just sub in $t = 120$ into $N(t)$.
- iii. Many errors. We are finding a rate of change so we need to use $N'(t)$. Some students made mistakes differentiating the exponential function.
- iv. Mostly done well. Some students made simple errors.

Question 7

- a) (i) 1 mark given for a correct chain rule expression. The second mark was for the correct answer either given as a negative or in words "at a decreasing rate".

(ii) Quite poorly done. Two ways to do it were as per solutions or by expressing b in terms of l and substituting this into the perimeter formula. You could then write another chain rule expression and sub in $l=8$ to evaluate the answer of 2.5. See below.

$$\begin{aligned}
 P &= 2(l + b) \\
 P &= 2\left(l + \frac{48}{l}\right) \\
 \frac{dP}{dt} &= \frac{dP}{dl} \times \frac{dl}{dt} \\
 \text{Now } \frac{dP}{dl} &= 2\left(1 - \frac{48}{l^2}\right) \text{ and sub in } l = 8 \\
 \text{So } \frac{dP}{dt} &= 2\left(1 - \frac{48}{64}\right) \times 5 \\
 &= 2.5
 \end{aligned}$$

- b) OH the pain! AN ODD POLYNOMIAL ALWAYS PASSES THROUGH THE ORIGIN AND HAS SYMMETRICAL ZEROS ACROSS THE ORIGIN. NEVER EVER FORGET THIS IN THE FUTURE. This means the 5 zeros are 0, + and -1, + and -3. The polynomial can hence be expressed as $ax(x-1)(x+1)(x-3)(x+3)$ [question did not state it is monic so you needed the coefficient of a]. Sub in the point (2,60) to evaluate a and finally EXPAND the brackets as this is what question asked for.
- c) 4 solutions in total. Only one mark given for two, two marks for 3 and 3 marks for 4. Trickiest part was solving $\sin 2x = 1$. $2x = \pi/2$ and $5\pi/2$ because you have to solve $2x$ over $0 < x < 4\pi$ and then divide your answers by 2 to bring the answer of $5\pi/2$ back into the domain of $0 < x < 2\pi$.

Question 8

a) i. neatness and students remembering to show evidence that the section of the graph that stayed the same as the original was included on their diagram.

ii. only the positive section of $f(x)$ can be square rooted and then the section $0 \leq y < 1$ will create answers above the original graph and the same value at $y = 1$ as $\sqrt{1} = 1$.

b) poor use of index laws as many students incorrectly manipulated 2^{-t} the quickest way was actually to do:

$y = 2^t \rightarrow \frac{1}{y} = 2^{-t}$ then sub this into the $x =$ to create the rule for three marks it was necessary to have the answer in $y =$ format.

c) students failed to see the use of chain rule $\left[\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \right]$ to find dy/dx here and many did not find the co-ordinates of the point. I tried to look for those things when awarding part marks.

Question 9

a) Mainly well done.

b) i. well done

ii. Well done

iii. Many students failed to recognise $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ and then having to substitute correctly using the appropriate answers found in part i. and ii.

iv. Many errors here please have a good look at the solutions to this question.

c) 3 critical points to consider here $x = -2$, $x = 0$ and $x = 2$ then testing these points to see which satisfy the equation – see solutions. Note that x can not equal to any of these points so $\geq$ or $\leq$ are incorrect.

Question 10

a) This is a show question. Please show all your working.

Please refer to the reference sheet if you forgot how to find the area of a sector.

b) Acceleration is the derivative of the velocity function. Also, please express your answer in the simplest form.

c) Many people could not differentiate $4x^3 - 27x + k$ correctly.

d) There were many non-attempts, perhaps time pressure

Question 11

Many students were clearly running out of time for this question, so the comments need to be viewed in this context.

- a) i) This is a routine concept in rates. It is almost a trivial show, to gain the mark students needed either the line $Ae^{-kt} = T - 26$ before substituting or needed to write the derivative as $\frac{dT}{dt} = -k(Ae^{-kt} + 26 - 26)$
- ii) Again, a routine extension 1 rates question. Of concern was the fact that a non-trivial number of students did not successfully find the value of A
- b) i) Poorly done. Much to the distress of the marker many students did not recognise it is a parabola. There were several creative improvisations that did not score any marks. For full marks you needed most labelled points, a correct 'general' shape (and it was marked generously) and the restriction. For the record, it is an Advanced and even a year 10 concept really, given it is written in 'completed square form' the vertex can virtually be read of the equation as (4,3). The intercepts can be found easily, and the coefficient of the x^2 term is negative.
- ii) An early year 11 concept. Many students seemed to get lost after swapping the x and y . The idea is to then make y the subject, so you need to move the 4 across to LHS and then take a square root. Expanding the RHS is a dead end as we end up with a y and a y^2 term which is the opposite of what we want y on its own so it can be made the subject. For the final mark you have the you need to first arrive at $3 \pm 3\sqrt{4-x}$ (or similar) and then remove the $+$ case by considering the domain/range of the inverse (D/R swapped for inverse)
- iii) **see comment for 7b) DON'T FORGET THIS** you need to be aware of this important concept of functions and inverses, as it often crops up in harder functions questions. The intersection of $f(x) = f^{-1}(x)$ also coincides with the line $y = x$. So $f(x) = f^{-1}(x)$ is a mistake and leads to a dead end. Successful students recognised this concept and started with $f(x) = x$, i.e. $4 - \left(1 - \frac{x}{3}\right)^2 = x$, after this step you just then expand and gather terms to side and solve quadratic using the quadratic formula. Due to domain/range instructions there is only one point of intersection, and they need to consider the domain and range when selecting the point. Full marks were awarded for finding the correct x coordinate, however the question did ask for a point. Again, you need to use to fact that the point also lies on $y = x$, so just y ordinate is the same as the x ordinate. Several students plugged in the x into $f(x)$ and wasted time and up to one page of flawed algebra to arrive at an incorrect y ordinate